RTS3917N
(Part Number: RTS3917N-GR)

3 Mega Pixel H.265/H.264 IP Camera SoC

DATASHEET

Rev 1.1
29th Apr. 2022

Realtek Semiconductor Corp.
No. 2, Innovation Road II, Hsinchu Science Park, Hsinchu 300, Taiwan
Tel.: +886-3-578-0211. Fax: +886-3-577-6047
www.realtek.com

USING THIS DOCUMENT

This document is intended for the hardware and software engineer's general information on the Realtek RTS3917N IP camera SoC.

Though every effort has been made to ensure that this document is current and accurate, more information may have become available subsequent to the production of this guide. In that event, please contact your Realtek representative for additional information that may help in the development process.

REVISION HISTORY

Revision	Release Date	Summary
1.0	2022/02/17	First release
1.1	2022/04/29	Update GPIO pin definition, power pin naming, pin function description, and features of audio, SAR-ADC

Table of Contents

1. GENERAL DESCRIPTION 1
1.1. INTRODUCTION 1
1.1.1. Application Scenarios 1
1.2. OVERVIEW 1
2. ARCHITECTURE 2
3. FEATURES 3
4. PIN ASSIGNMENT 5
4.1. QFN88 5
4.2. PACKAGE AND VERSION IDENTIFICATION 6
5. PIN DESCRIPTIONS 7
5.1. USB TRANSCEIVER INTERFACE 7
5.2. SYSTEM INTERFACE 7
5.3. SPI SERIAL FLASH INTERFACE 8
5.4. SD/EMMC INTERFACE 8
5.5. UART INTERFACE 9
5.6. SAR-ADC INTERFACE 10
5.7. CMOS IMAGE SENSOR INTERFACE 10
5.8. PWM INTERFACE 11
5.9. AUDIO INTERFACE 11
5.10. ETHERNET INTERFACE 11
5.11. GPIO 12
5.12. POWER/GROUND 12
6. PERIPHERAL 13
6.1. I2C 13
6.1.1. Features 13
6.1.2. I2C timing 13
6.2. GPIO 15
6.2.1. Features 15
6.2.2. Operation Flow 16
6.3. UART 16
6.3.1. Features 16
6.3.2. Function Description 16
6.4. SAR-ADC 17
6.4.1. Features 17
6.5. USB HOST 18
6.6. THERMAL SENSOR 18
6.6.1. Features 18
7. AUDIO 19
7.1. OVERVIEW 19
7.2. FEATURES 19
8. MEMORY & STORAGE 20
8.1. DDR CONTROLLER 20
8.2. SPI CONTROLLER 20
8.2.1. Overview 20
8.2.2. Features 20
8.3. SDIO 21

8.3.1. Features *21*

9. AES **22**

9.1. *GENERAL DESCRIPTION* 22
9.2. *FEATURES* 22

10. ELECTRICAL CHARACTERISTICS **23**

10.1. *DC CHARACTERISTICS* 23
10.1.1. Absolute Maximum Ratings *23*
10.2. *POWER ON SEQUENCE* 23

11. APPLICATION CIRCUITS **24**

12. PACKAGE MECHANICAL DIMENSIONS **25**

12.1. *QFN88 DIMENSION* 25

13. ORDERING INFORMATION **27**

List of Tables

TABLE 5-1 I/O TYPE DESCRIPTION......7
TABLE 5-2 USB TRANSCEIVER INTERFACE......7
TABLE 5-3 SYSTEM INTERFACE......7
TABLE 5-4 SERIAL FLASH INTERFACE......8
TABLE 5-5 SD/EMMC INTERFACE......8
TABLE 5-6 UART INTERFACE......9
TABLE 5-7 SAR-ADC INTERFACE......10
TABLE 5-8 CMOS IMAGE SENSOR INTERFACE......10
TABLE 5-9 PWM INTERFACE......11
TABLE 5-10 AUDIO INTERFACE......11
TABLE 5-11 ETHERNET INTERFACE......11
TABLE 5-12 GPIO INTERFACE......12
TABLE 5-13 POWER AND GROUND PINS......12
TABLE 6-1. CHARACTERISTICS OF THE SDA AND SCL I/O STAGES FOR F/S-MODE I^2C-BUS DEVICES......13
TABLE 6-2. CHARACTERISTICS OF THE SDA AND SCL BUS LINES FOR F/S-MODE I^2C-BUS DEVICES(1)......14
TABLE 10-1 ABSOLUTE MAXIMUM RATINGS......23
TABLE 10-2 POWER REQUIREMENTS......24
TABLE 13-1. ORDERING INFORMATION......27

List of Figures

FIGURE 1-1 IP CAMERA SOLUTION 1
FIGURE 2-1 ARCHITECTURE DIAGRAM 2
FIGURE 4-1 RTS3917N QFN88 PIN DEFINE 5
FIGURE 6-1 DEFINITION OF TIMING FOR F/S-MODE DEVICES ON THE I^2C-BUS. 13
FIGURE 6-2 SERIAL DATA FORMAT 16
FIGURE 6-3 AUTO FLOW CONTROL DIAGRAM UART 17
FIGURE 10-1 RTS3917N POWER ON SEQUENCE 23

1. General Description

1.1. *Introduction*

1.1.1. *Application Scenarios*

As a professional system-on-chip (SoC), the RTS3917N is designed for the IP camera. Based on application requirements, it encodes multiple streams in H.265/H.264 format and provides superior image signal processing (ISP) performance, high encoded video quality, and a high-performance intelligent acceleration engine. With these features, it meets customers' requirements for product functions, performance, and image quality, and helps customers significantly reduce the engineering bill of materials costs. The following describes typical 3M IP camera solutions.

1.2. *Overview*

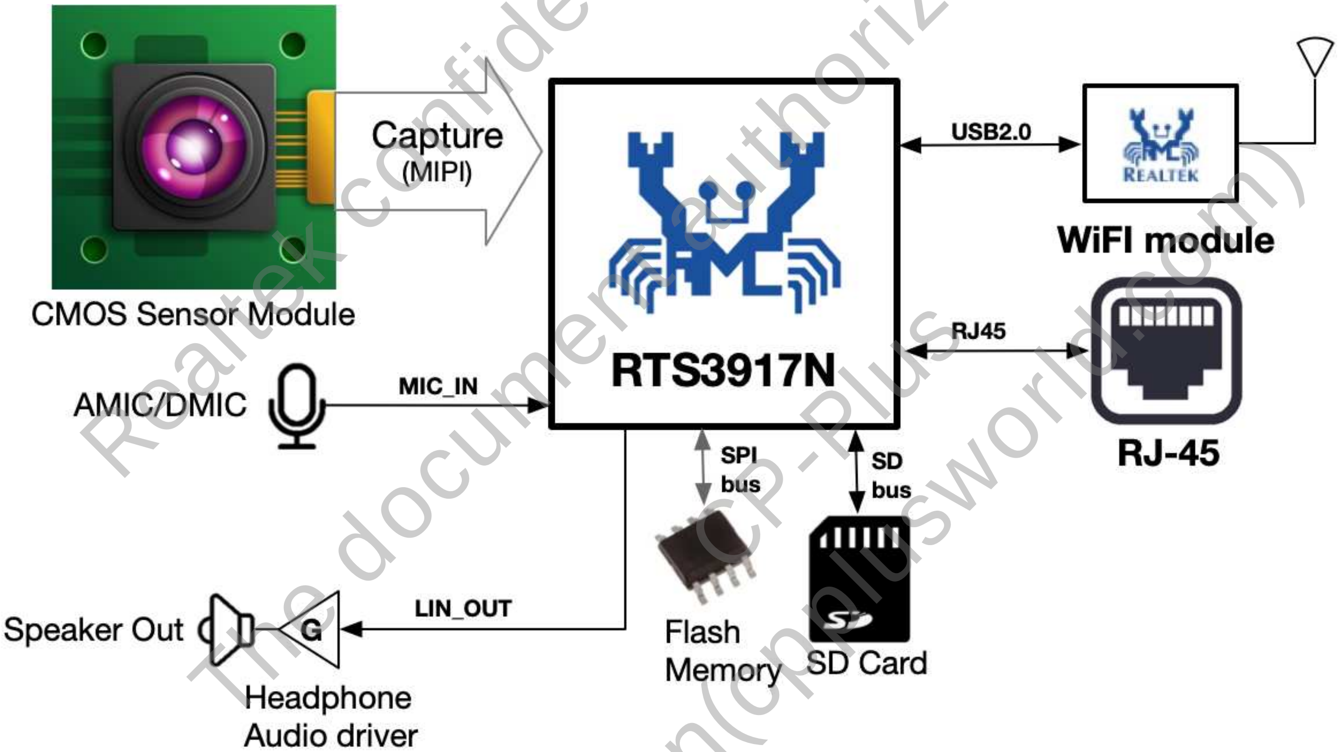


Figure 1-1 IP Camera solution

2. Architecture

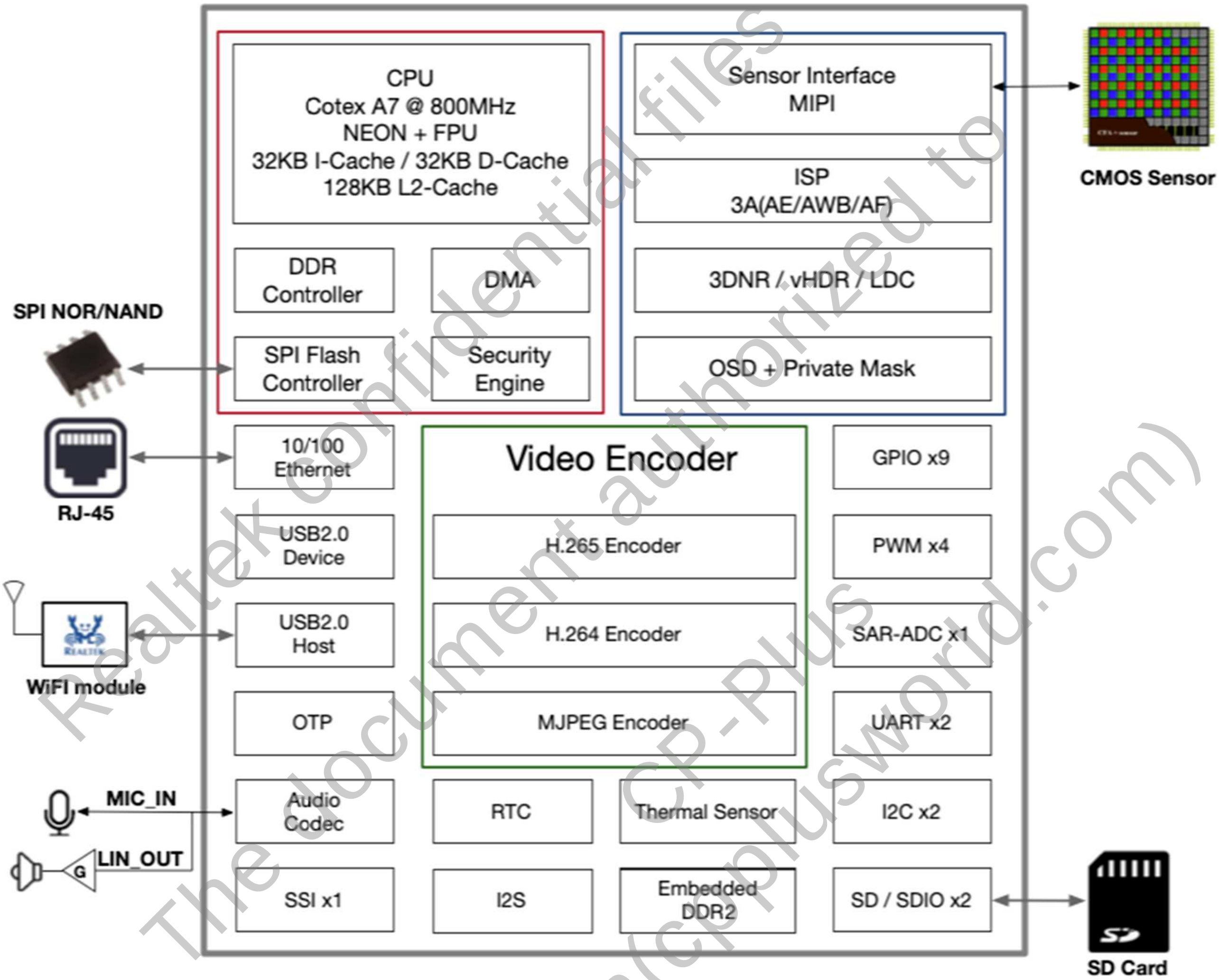


Figure 2-1 Architecture Diagram

3. Features

Central Processing Unit

- ARM Cortex-A7
- CPU Clock Frequency 800MHz
- Neon and FPU
- 128KB L2 Cache/32KB I-Cache/ 32KB D-Cache

Integrated Memory

- Support 16-bit DDR2
- Integrated 512Mbits DDR2

SPI NOR Flash

- Support single/dual/quad channel read/write mode
- Support SPI NOR flash
- Support 4 byte address mode for 32MB flash

SPI NAND Flash

- Support single/dual/quad channel read/write mode
- Support up to 2Gbits SPI NAND flash with 2KB page size

Security Engine

- Supports AES-128/192/256-ECB/ CBC/CTR/GCM/CCM
- Supports RSA-512/1024/2048/3072
- Supports SHA256/HMAC-SHA256
- Supports TRNG
- AES data block length 128bits
- Support multi-buffer encryption/decryption
- Support inter-buffer data splicing
- Support auto-padding
- OTP (One Time Programmable) 16kbits

Ethernet

- Embedded 10/100 Mac and Physical Layer

Wireless

- Support USB2.0 Wi-Fi module
- Support SDIO Wi-Fi module

Sensor interface

- MIPI 2-lane interface up to 12-bit Raw data
- HCLK support 24MHz/27MHz or 54MHz/37.125MHz or 74.25MHz

Image Signal Processor

- Support individual BLC (Black Level Cancellation) or OBC (Optical Black Cancellation)
- Individual LSC (Lens Shading Correction) for Gr/R/B/Gb channels
- LSC supports both of circle and grid compensation
- On-the-fly DPC (defect pixel compensation)
- Support individual 3A (AE/AWB/AF) statistics
- Advanced color interpolation (de-mosaicking) with false color reduction
- Programmable R/G/B Gamma
- Support WDR engine and auto WDR control
- 2 exposures vHDR (Video HDR)

- Support Auto Dynamic Range Compression
- Support de-haze engine
- Flexible CSC (Color Space Conversion) with default of BT.709 standard
- Advance EEH (Edge Enhance) with improvement of detail and sharpness
- Capability of horizontal LDC (Lens Distortion Correction)
- Whole new designed 3DNR (3D Noise Reduction) with suppression of noise vibration
- Support private mask
- Support zoom out with 5 channels
- OSD supports 6 areas and monochrome pictures with 5 channels
- OSD supports alpha-blending process
- Support ISP tuning tools for the PC through Ethernet/Wi-Fi

H.265/H.264 Encoder

- Support H.265/H.264 encoder
- Support H.265/H.264 MBRC
- Support H.265/H.264 CBR/VBR/CVBR Rate control mode
- Support H.265/H.264 ROI Map
- Support video rotates
- Maximum H.265/H.264 resolution is 3M
- H.264 encoder support Base Line/Main Profile /High Profile
- H.265 encoder support Main Profile
- Supports up to quarter-pixel
- Supports frame level and MB level rate control
- Supports ROI encoding with custom QP map

JPEG Encoder

- Supports JPEG baseline encoding
- Supports YUV422 or YUV420 formats
- Supports max. 3M with 25fps encoding

Video Streaming

- Support H.265/H.264 2-stream : 3M@25fps + VGA@25fps
- H.265/H.264 stream support CBR/VBR/CVBR
- Support MJPEG 3M@25fps

Audio Processor

- Build-in one 24-bit audio codec
- Support mono ADC/DAC
- Support DMIC input
- Support I2S
- Support sample rate: 192kHz/ 176.4kHz/ 96kHz/ 88.2kHz/ 48kHz/ 44.1kHz/ 32kHz/ 24 kHz/ 22.05KHz/ 16 kHz/ 11.025 KHZ/ 8kHz
- Support G.711/G.726/AAC encoding
- Software Acoustic echo cancellation

Peripheral

- GPIO x9, PWM x4, SARADC 1-channel
- UART x2
- I2C interface, SSI interface
- USB Host 2.0 & USB Device 2.0
- SD 2.0 x2 or SD 2.0 x1 & eMMC x1
- RTC
- Embedded thermal sensor

4. Pin Assignment

4.1. *QFN88*

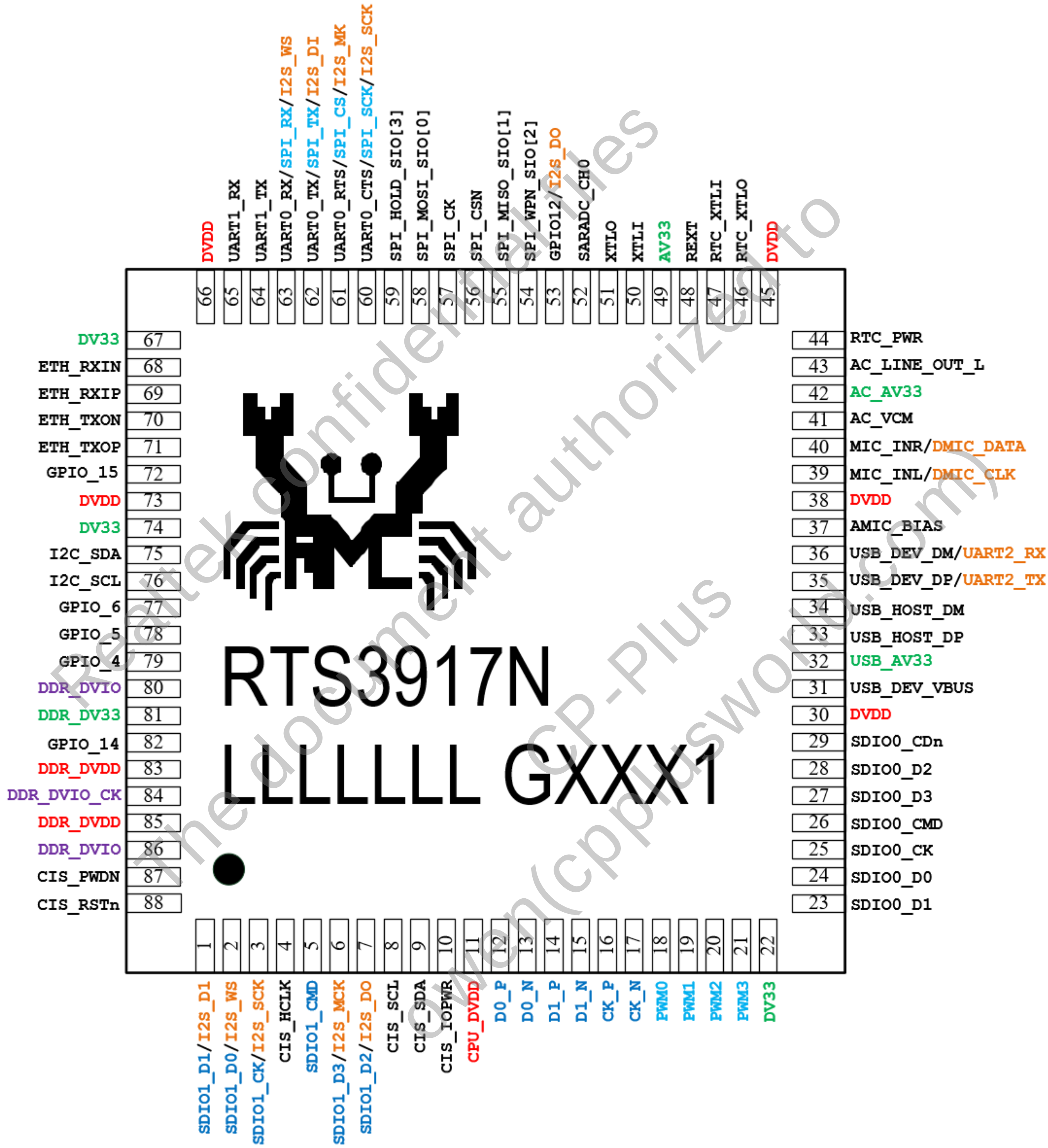


Figure 4-1 RTS3917N QFN88 pin define

4.2. Package and Version Identification

Green package is indicated by a ‘G’ in the location marked ‘T’. The silicon version number is shown in the location marked ‘V’.

5. Pin Descriptions

Table 5-1 I/O Type Description

I/O Type	Description
I	Input
O	Output
IH	Input with internal pull-up 200K
IL	Input with internal pull-down 200K
IO	Input / Output
IOH	Input/Output with internal pull-up 200K
IOL	Input/Output with internal pull-down 200K
IOSH	Input/Output with Schmitt trigger
IO-U	USB related IO
CLK	Clock related IO
PWR-O	Power output pin
PWR-I	Power input pin
GND	Ground related pin

5.1. *USB Transceiver Interface*

Table 5-2 USB Transceiver Interface

Name	Type	Pin No.	Description
USB_DEV_VBUS	I/O	31	Function 1: For detect the USB host device plug in to our USB device port Function 2: GPIO
USB_AV33	PWR-I	32	USB BUS 3.3V input
USB_HOST_DP	IO-U	33	Function 1: USB host DP, downstream port Function 2: GPIO
USB_HOST_DM	IO-U	34	Function 1: USB host DM, downstream port Function 2: GPIO
USB_DEV_DP	IO-U	35	Function 1: USB device DP, upstream port Function 2: UART2_TXD Function 3: GPIO
USB_DEV_DM	IO-U	36	Function 1: USB device DM, upstream port Function 2: UART2_RXD Function 3: GPIO
			Total: 6 Pins

5.2. *System Interface*

Table 5-3 System Interface

Name	Type	Pin No.	Description
RTC_PWR	PWR-I	44	It usually connects to button cell battery for RTC power
RTC_XTLO	O	46	32.768KHz Crystal output

RTC_XTLI	I	47	32.768KHz Crystal input
REXT	I	48	External 12KOhm 1% resistor to ground for internal reference
XTLI	I	50	25M Crystal input
XTLO	O	51	25M Crystal output
I2C_SDA	I/O	75	Function 1: I2C_SDA Function 2: GPIO Function 3: ETH_LED1 Function 4: MDIO_DAT Function 5: SPI_CLK
I2C_SCL	I/O	76	Function 1: I2C_SCL Function 2: GPIO Function 3: ETH_LED0 Function 4: MDIO_CLK Function 5: SPI_CS
			Total: 8 Pins

5.3. *SPI Serial Flash Interface*

Table 5-4 Serial Flash Interface

Name	Type	Pin No.	Description
SPI_WPN_SIO[2]	IOH	54	Function 1: Write Protect/Serial Function 2: Data Input & Output 2 Function 3: GPIO
SPI_MISO_SIO[1]	IOH	55	Serial data in/Serial Data Input & Output 1
SPI_CSn	IOH	56	SPI Flash Chip Select
SPI_CLK	IOH	57	Clock signal
SPI_MOSI_SIO[0]	IOH	58	Serial data out/Serial Data Input & Output 0
SPI_HOLD_SIO[3]	IOH	59	Function 1: Hold/Serial Data Input & Output 3 Function 2: GPIO Function 3: PWM3
			Total: 6 Pins

5.4. *SD/eMMC Interface*

Table 5-5 SD/eMMC Interface

Name	Type	Pin No.	Description
SD0_D[1]	I/O	23	Function 1: SD0 data 1 or eMMC data 1 Function 2: GPIO
SD0_D[0]	I/O	24	Function 1: SD0 data 0 or eMMC data 0 Function 2: GPIO
SD0_CK	O	25	Function 1: SD0 clock pin or eMMC CLK Function 2: GPIO
SD0_CMD	I/O	26	Function 1: SD0 Command pin or eMMC CMD Function 2: GPIO
SD0_D[3]	I/O	27	Function 1: SD0 Data 3 or eMMC data 3 Function 2: GPIO

SD0_D[2]	I/O	28	Function 1: SD0 Data 2 or eMMC data 2 Function 2: GPIO
SD0_CDn	I/O	29	Function 1: SD0 card detect Function 2: GPIO
SD1_D[0]	I/O	2	Function 1: GPIO Function 2: SDWIFI_DATA[0] Function 3: I2S_WS
SD1_CK	I/O	3	Function 1: GPIO Function 2: SDWIFI_CLK Function 3: I2S_SCLK
SD1_CMD	I/O	5	Function 1: GPIO Function 2: SDWIFI_CMD
SD1_D[3]	I/O	6	Function 1: GPIO Function 2: SDWIFI_DATA[3] Function 3: I2S_MCK
SD1_D[2]	I/O	7	Function 1: GPIO Function 2: SDWIFI_DATA[2] Function 3: I2S_DATA_O
			Total: 12 Pins

5.5. *UART Interface*

Table 5-6 UART Interface

Name	Type	Pin No.	Description
UART0_CTS	I/O	60	Function 1: GPIO Function 2: UART0 CTS Function 3: JTAG_TCK Function 4: SSI_SCLK Function 5: SWDCLK Function 6: I2S_SCLK
UART0_RTS	I/O	61	Function 1: GPIO Function 2: UART0 RTS Function 3: JTAG_TMS Function 4: SSI_CS_0 Function 5: SWDIO Function 6: I2S_MCLK
UART0_TX	I/O	62	Function 1: GPIO Function 2: UART0 TX Function 3: JTAG_TDI Function 4: SSI_TX Function 5: SPI_MOSI / SIO[0] Function 6: I2S_DATA_I

UART0_RX	I/O	63	Function 1: GPIO Function 2: UART0 RX Function 3: JTAG_TRST Function 4: SSI_RX Function 5: SPI_MISO / SIO[1] Function 6: I2S_WS
UART1_TX	I/O	64	Function 1: UART1 TX for console debug port Function 2: GPIO
UART1_RX	I/O	65	Function 1: UART1 RX for console debug port Function 2: GPIO
			Total: 6 Pins

5.6. *SAR-ADC Interface*

Table 5-7 SAR-ADC Interface

Name	Type	Pin No.	Description
SARADC_CH[0]	I	52	Function 1: SAR-ADC channel 0 Function 2: GPIO
			Total: 1 Pin

5.7. *CMOS Image Sensor Interface*

Table 5-8 CMOS Image Sensor Interface

Name	Type	Pin No.	Description
CIS_HCLK	O	4	System master clock of sensor
CIS_SCL	I/O	8	Function 1: XB2 I2C1 Bus CLOCK Function 2: GPIO
CIS_SDA	I/O	9	Function 1: XB2 I2C1 Bus DATA Function 2: GPIO
CIS_IOPWR	PWR-I	10	Power input with the same IO voltage as CMOS image sensor
D0_P	I	12	Function 1: MIPI_RX interface:D0_P Function 2: GPI
D0_N	I	13	Function 1: MIPI_RX interface:D0_N Function 2: GPI
D1_P	I	14	Function 1: MIPI_RX interface:D1_P Function 2: GPI
D1_N	I	15	Function 1: MIPI_RX interface:D1_N Function 2: GPI
CK_P	I	16	Function 1: MIPI_RX interface:CK_P
CK_N	I	17	Function 1: MIPI_RX interface:CK_N
			Total: 10 Pins

5.8. *PWM Interface*

Table 5-9 PWM Interface

Name	Type	Pin No.	Description
PWM[0]	I/O	18	Function 1: GPIO Function 2: PWM0 OUTPUT
PWM[1]	I/O	19	Function 1: GPIO Function 2: PWM1 OUTPUT
PWM[2]	I/O	20	Function 1: GPIO Function 2: PWM2 Output
PWM[3]	I/O	21	Function 1: GPIO Function 2: PWM3 Output
			Total: 4 Pins

5.9. *Audio Interface*

Table 5-10 Audio Interface

Name	Type	Pin No.	Description
AMIC_BIAS	PWR-O	37	LDO output for AMIC bias
MIC_INL	I/O	39	Function 1: GPIO Function 2: AMIC_IN Left Channel Function 3: DMIC1_CLK
MIC_INR	I/O	40	Function 1: GPIO Function 2: AMIC_IN Right Channel Function 3: DMIC1_DAT
AC_VCM	PWR-O	41	Common mode voltage, External 1uF to GND
AC_AV33	PWR-I	42	For AMIC power 3.3V input
AC_LINE_OUT_L	I/O	43	Function 1: GPIO Function 2: Line out left channel
			Total: 6 Pins

5.10. *Ethernet Interface*

Table 5-11 Ethernet Interface

Name	Type	Pin No.	Description
ETN_RXIN	I/O	68	100-ohm differential receive negative input ,support cross-over mode
ETN_RXIP	I/O	69	100-ohm differential receive positive input, support cross-over mode
ETN_TXON	I/O	70	100-ohm differential transmit negative output, support cross-over mode
ETN_TXOP	I/O	71	100-ohm differential transmit positive output, support cross-over mode
			Total: 4 Pins

5.11. *GPIO*

Table 5-12 GPIO Interface

Name	Type	Pin No.	Description
GPIO_4	I/O	79	Function 1: GPIO Function 2: Work Mode[0]
GPIO_5	I/O	78	Function 1: GPIO Function 2: Work Mode[1]
GPIO_6	I/O	77	Function 1: GPIO Function 2: Work Mode[2]
GPIO_8	I/O	1	Function 1: GPIO Function 2: SDWIFI_DATA[1] Function 3: I2S_DATA_I
GPIO_9	I/O	88	Function 1: GPIO
GPIO_10	I/O	87	Function 1: GPIO
GPIO_12	I/O	53	Function 1: GPIO Function 2: JTAG_TDO Function 3: I2S_DATA_O Function 4: SSI_CS_1
GPIO_14	I/O	82	Function 1: GPIO
GPIO_15	I/O	72	Function 1: GPIO
			Total: 9 Pin

5.12. *Power/Ground*

Table 5-13 Power and Ground pins

Name	Type	Pin No.	Description
CPU_DVDD	PWR-I	11	CPU power 0.9V input
DVDD	PWR-I	30, 38, 45, 66, 73	Core power 0.9V input
DV33	PWR-I	22, 67, 74	IO power 3.3V input
AV33	PWR-I	49	System 3.3V Analog power input
DDR_DVIO	PWR-I	80, 86	DDR Controller IO power 1.5V input
DDR_DV33	PWR-I	81	DDR PHY PLL 3.3V power input
DDR_DVDD	PWR-I	83, 85	Power for DDR controller 0.9V input
DDR_DVIO _CK	PWR-I	84	DDR Controller PLL 1.5V input
DGND	GND	EPAD	Ground
			Total: 17 Pins

6. Peripheral

6.1. *I2C*

The inter-integrated circuit (I2C) interface is a simple two-wire bus with a software-defined protocol for system control, which is used in temperature sensors and voltage level translators to EEPROMs, general-purpose I/O, A/D and D/A converters, CODECs, and many types of microprocessors.

The I2C bus is a two-wire serial interface, consisting of a serial data line (SDA) and a serial clock (SCL). These wires carry information between the devices connected to the bus. Each device is recognized by a unique address and can operate as either a “transmitter” or “receiver,” depending on the function of the device. Devices can also be considered as masters or slaves when performing data transfers. A master is a device that initiates a data transfer on the bus and generates the clock signals to permit that transfer. At that time, any device addressed is considered a slave.

6.1.1. *Features*

The I2C module has the following features:

- Supports clock synchronization and bit and byte waiting.
- Supports interrupts or polling.
- Supports 7-bit standard address and 10-bit extended address.
- Supports standard mode (100 kbit/s) and high-speed mode (400 kbit/s).
- Supports general call and start byte.
- Filters the received serial data (SDA) and serial clock (SCL) signals

6.1.2. *I2C timing*

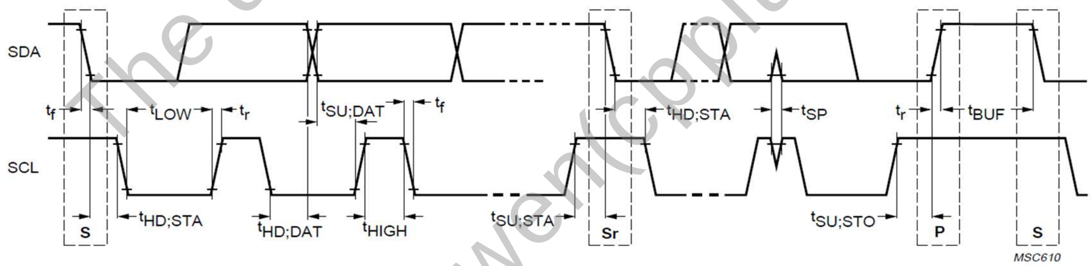


Figure 6-1 Definition of timing for F/S-mode devices on the I²C-bus.

Table 6-1. Characteristics of the SDA and SCL I/O stages for F/S-mode I²C-bus devices

PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
LOW level input voltage: fixed input levels	VIL	−0.5 −0.5	1.5 0.3VDD	n/a −0.5	n/a 0.3VDD(1)	V V

VDD-related input levels						
HIGH level input voltage: fixed input levels VDD-related input levels	VIH	3.0 0.7VDD	(2) (2)	n/a 0.7VDD(1)	n/a(2)	V V
Hysteresis of Schmitt trigger inputs: VDD > 2 V VDD < 2 V	Vhys	n/a n/a	n/a n/a	0.05VDD 0.1VDD	– –	V V
LOW level output voltage (open drain or open collector) at 3 mA sink current: VDD > 2 V VDD < 2 V	VOL1 VOL3	0 n/a	0.4 n/a	0 0	0.4 0.2VDD	V V
Output fall time from VIHmin to VILmax with a bus capacitance from 10 pF to 400 pF	tof	–	250(4)	20 + 0.1Cb(3)	250(4)	ns
Pulse width of spikes which must be suppressed by the input filter	tSP	n/a	n/a	0	50	ns
Input current each I/O pin with an input voltage between 0.1VDD and 0.9VDDmax	Ii	−10	10	−10(5)	10(5)	µA
Capacitance for each I/O pin	Ci	–	10	–	10	pF

Notes
1. Devices that use non-standard supply voltages that do not conform to the intended I^{2}C-bus system levels must relate their input levels to the V_{DD} voltage to which the pull-up resistors R_p are connected.
2. Maximum $V_{IH} = V_{DDmax} + 0.5$ V.
3. C_b = capacitance of one bus line in pF.
4. The maximum t_f for the SDA and SCL bus lines quoted in Table 5 (300 ns) is longer than the specified maximum t_{of} for the output stages (250 ns). This allows series protection resistors (R_s) to be connected between the SDA/SCL pins and the SDA/SCL bus lines as shown in Fig.36 without exceeding the maximum specified t_f.
5. I/O pins of Fast-mode devices must not obstruct the SDA and SCL lines if V_{DD} is switched off.
6. n/a = not applicable

Table 6-2. Characteristics of the SDA and SCL bus lines for F/S-mode I^{2}C-bus devices(1)

PARAMETER	SYMBOL	STANDARD-MODE		FAST-MODE		UNIT
		MIN.	MAX.	MIN.	MAX.	
SCL clock frequency	f_{SCL}	0	100	0	400	kHz
Hold time (repeated) START condition. After this period, the first clock pulse is generated	$t_{HD;STA}$	4.0	–	0.6	–	µs
LOW period of the SCL clock	t_{LOW}	4.7	–		–	µs
HIGH period of the SCL clock	t_{HIGH}	4.0	–	0.6	–	µs
Set-up time for a repeated START	$t_{SU;STA}$	4.7	–	0.6	–	µs

condition						
Data hold time: for CBUS compatible masters for I²C-bus devices	$t_{HD;DAT}$	5.0 0(2)	– 3.45(3)	– 0(2)	– 0.9(3)	µs µs
Data set-up time	$t_{SU;DAT}$	250	–	100(4)	–	ns
Rise time of both SDA and SCL signals	t_r	–	1000	$20 + 0.1C_b$(5)	300	ns
Fall time of both SDA and SCL signals	t_f	–	300	$20 + 0.1C_b$(5)	300	ns
Set-up time for STOP condition	$t_{SU;STO}$	4.0	–	0.6	–	µs
Bus free time between a STOP and START condition	t_{BUF}	4.7	–	1.3	–	µs
Capacitive load for each bus line	C_b	–	400	–	400	pF
Noise margin at the LOW level for each connected device (including hysteresis)	V_{nL}	$0.1V_{DD}$	–	$0.1V_{DD}$	–	V
Noise margin at the HIGH level for each connected device (including hysteresis)	V_{nH}	$0.2V_{DD}$	–	$0.2V_{DD}$	–	V

Notes

1. All values referred to V_{IHmin} and V_{ILmax} levels (see Table 6-1. Characteristics of the SDA and SCL I/O stages for F/S-mode I2C-bus devices).
2. A device must internally provide a hold time of at least 300 ns for the SDA signal (referred to the V_{IHmin} of the SCL signal) to bridge the undefined region of the falling edge of SCL.
3. The maximum $t_{HD;DAT}$ has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal.
4. A Fast-mode I²C-bus device can be used in a Standard-mode I²C-bus system, but the requirement $t_{SU;DAT} \geq 250$ ns must then be met. This will automatically be the case if the device does not stretch the LOW period of the SCL signal. If such a device does stretch the LOW period of the SCL signal, it must output the next data bit to the SDA line $t_{r\,max} + t_{SU;DAT} = 1000 + 250 = 1250$ ns (according to the Standard-mode I²C-bus specification) before the SCL line is released.
5. C_b = total capacitance of one bus line in pF. If mixed with Hs-mode devices, faster fall-times according to Table 5-4 are allowed.
6. n/a = not applicable

6.2. *GPIO*

A "General Purpose Input/Output" (GPIO) is a flexible software-controlled digital signal. They are provided from many kinds of chip, represents a bit connected to a particular pin, or "ball" on Ball Grid Array (BGA) packages. Board schematics show the external hardware GPIOs connections.

6.2.1. *Features*

RTS3917N contains some GPIO and multi-functional input/output port pins. The GPIO has the following features:

- Can controls interruption or boot up mode by the GPIO pins
- Multi-functional input/output port pins can be configured as USB, PWM, UART and Audio functions

6.2.2. *Operation Flow*

- Set relevant bit in XB2_SHARE_GPIO_SELECT to select GPIO function
- Set relevant bit in register XB2_SHARE_GPIO_OE to select direction
- Set relevant bit in register XB2_SHARE_GPIO_O to select output level, or read to get input level
- Set relevant bit in register XB2_SHARE_GPIO_INT_EN to enable interrupt
- Read relevant bit in register XB2_SHARE_GPIO_INT_FLAG to know which interrupt happened

6.3. *UART*

6.3.1. *Features*

Three UART interfaces are provided in IPCAM SoC. UART provides a serial communication interface with a computer. Only one UART interfaces support auto flow control with CTS, RTS. The interface support standard of 16550A. The UART interface has the following features:

- 32-byte FIFO for both TX and RX to reduce the number of interrupts sent to the host processor
- Adds or deletes standard asynchronous communication bits (start, stop, and parity) to or from the serial Data
- Programmable baud rate generator divides UART clock by DL (divisor latch) and generate the 16x baud rate clock. The UART clock is 24MHz.
- Fully programmable serial interface characteristics.
- 5, 6, 7 or 8-bit characters.
- Even, odd or no-parity bit generation and detection
- 1, 1.5, or 2 stop bits generation

6.3.2. *Function Description*

UART transport protocol is as follows:

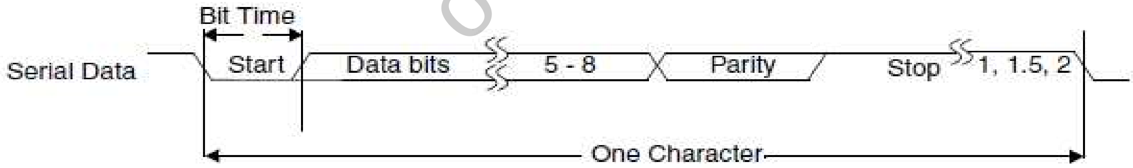


Figure 6-2 Serial Data Format

Auto flow control need to be supported by the sender and the receiver, connection diagram is as follows:

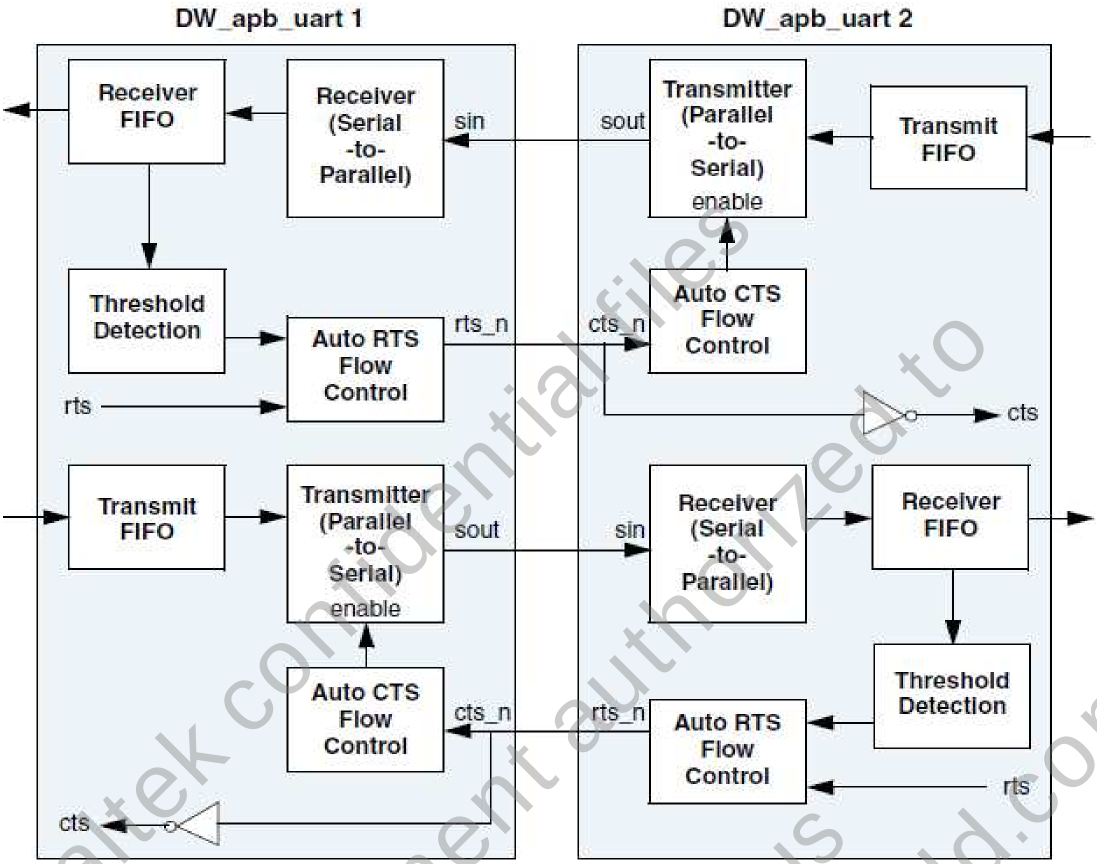


Figure 6-3 Auto Flow Control Diagram UART

6.4. *SAR-ADC*

6.4.1. *Features*

SAR-ADC is a type of analog-to-digital converter that converts a continuous analog waveform into a discrete digital. The IP Camera SOC has 1-channel, 8bit ADC

- 8 bit precision, FS = 1 ms/s, DNL = + 0.5 LSB
- Support 1 channel, use the same set of VREF, time division multiplexing
- Detection range of 0~3.3V, 0 corresponding to 0, 3.3V corresponding to 255

6.5. *USB Host*

The USB 2.0 Host Controller is fully compliant with the Universal Serial Bus Specification, Revision 1.1, and Enhanced Host Controller Interface Specification for Universal Serial Bus, Revision 2.0, and the openHCI: Open Host Controller Interface Specification for USB, Release 1.0a. The controller supports high speed, 480-Mbps transfers (40 times faster than USB 1.1 full-speed mode) using an EHCI Host Controller, as well as full and low speeds through one or more integrated OHCI Host Controllers.

- Design methodology supports full scan for testability
- All clock synchronization is handled within the controller
- Descriptor and data prefetching (configuration option)
- No bidirectional or three-state buses
- No level-sensitive latches
- Complies with Enhanced Host Controller Interface (EHCI) Specification, Version 1.0, and the Open Host Controller Interface (OHCI) Specification, Version 1.0a.
- Complies with the USB 2.0 Specification
- Supports ping and split transactions
- UTMI (legacy), UTMI+, ULPI, or HSIC interface to the PHY

6.6. *Thermal sensor*

6.6.1. *Features*

A temperature sensor is a device, typically, that provides for temperature measurement through an electrical signal.

- Temperature is read as a 19–bit digital value.
- User–definable temperature alarm settings, alarm interrupt notify.
- Converts temperature to digital word fast

7. Audio

7.1. *Overview*

Audio module is designed for transferring voice from local to remote through network or vice versa. In this chip, one embedded codec (ADC/DAC) is supported.

7.2. *Features*

The audio module has the following features:

- Build-in one audio codec
- Support ADC/DAC, DMIC input
- Support I2S
- Support 8/16/24 bit PCM data
- Support sample rate: 192kHz/ 176.4kHz/ 96kHz/ 88.2kHz/ 48kHz/ 44.1kHz/ 32kHz/ 24 kHz/ 22.05KHz/ 16 kHz/ 11.025 KHZ/ 8kHz

This module has the following analog features:

- Line in
  - THD+N: -87 dBFS
  - SNR: 92 dBFS
- Line out
  - THD+N: -85 dBFS
  - SNR: 97 dBFS

8. Memory & Storage

8.1. *DDR Controller*

There are one DDR SDRAM controllers in RTS3917N SoC. It is a slave device on AXI Bus system. It supports high-performance interface for DRAM data and peripheral interface for setting internal control registers of AXI_WRAP and PCTL. It provides an interface between AXI bus and DDR2/ DDR3 SDRAM. It has flowing features:

- Supports power-on initialization automatically
- Supports refresh DRAM automatically
- Supports pre-open/close DRAM page if pervious data access are on different bank
- Supports DRAM command and data pipeline issues
- Support 16-bit data width on internal DDR2 SDRAM interface
- Support 512Mb memory size.
- Support DDR2-1066
- Supports 4 internal banks for DDR2
- Burst length 16 byte
- IO power 1.5V

8.2. *SPI Controller*

8.2.1. *Overview*

SPI Flash Controller is used to communicate with SPI flash. To operate Flash it needs to set registers and transmit serial data to Flash. The serial steps call user mode (user_mode). These steps are complex and routine. Importantly the user mode operation is not efficient. For users operate the controller easily and efficient, it supports automatic mode (auto_mode) to access Flash as access memory.

8.2.2. *Features*

SPI Flash Controller has the following features:

- High performance communication with SPI flash - Support 2-channel data bit to transmit/ receive.
- Easy and efficient to operate SPIC- Support automatic mode to translate AXI memory-mapping read/write transfers into SPI flash access.
- Supports interrupts and interrupt masking – Master collision, transmit FIFO overflow, transmit FIFO empty, receive FIFO underflow, and receive FIFO overflow interrupts can be masked independently
- Supports 3 byte/4 byte address mode

- Programmable features:
  - SPI channel number – Control channel bits of serial transfer.
  - SPI Clock rate – Control the bit rate of serial transfer, up to 60MHz.
  - FIFO entry size- Support programmable upper bound of FIFO for profiling.
  - Flash command registers - Flexible to set the different command codes for different flash vendors in automatic mode.
  - Dummy cycles- Allow users to add dummy cycles in receiving data path for timing tuning or extra pipelining registers.
  - Flash address size- Define the size of Flash to enable the slave select output in automatic mode.

8.3. *SDIO*

8.3.1. *Features*

IPCAM supports SD card, eMMC and SDIO device.

- High Voltage SD Memory Card – Operating voltage range: 2.7-3.6 V
- Bus Speed Mode (using 4 parallel data lines)
  - Default Speed mode: 3.3V signaling, Frequency up to 25 MHz, up to 12.5 MB/sec
  - High Speed mode: 3.3V signaling, Frequency up to 50 MHz, up to 25 MB/sec
- 32/16 bits DMA data bus
- Supports asynchronous FIFO for receive operations
- Support CRC7 & CRC16

9. AES

9.1. *General Description*

The CIPHER module is used for encrypting and decrypting video stream and audio stream. It implements the AES (Advanced Encryption Standard) algorithm. The AES implementation conforms to FIPS197.

9.2. *Features*

- AES supports keys with the length of 128 bits (AES-128), 192 bits (AES-192) and 256 bits (AES-256).
- AES only supports data blocks with the length of 128 bits.
- AES supports the ECB, CBC, CTR, GCM and CCM modes, which conform to NIST sp800-38(a/c/d).
- The CIPHER supports multi-buffer encryption and decryption. That means the data can be stored in several buffers.

10. Electrical Characteristics

10.1. *DC Characteristics*

10.1.1. *Absolute Maximum Ratings*

Table 10-1 Absolute Maximum Ratings

Parameter	Symbol	Minimum	Typical	Maximum	Units
Power Supply Input					
For RTS3917N CORE	DVDD	0.85	0.9	0.945	V
For RTS3917N I/O	DV33	3.15	3.3	3.45	V
For RTS3917N DDR2	DDR_DVIO	1.475	1.5	1.575	V
Ambient Operating Temperature	Ta	0	-	+70	ºC
Junction Temperature	Tj	-20	-	+125	ºC
ESD (Electrostatic Discharge, Human Body Mode)					
	Susceptibility Voltage				
All Pins	TBD				

10.2. *Power On Sequence*

RTS3917N Power on sequence: Core (0.9V) ≥ DDR (1.5V) ≥ IO (3.3V)

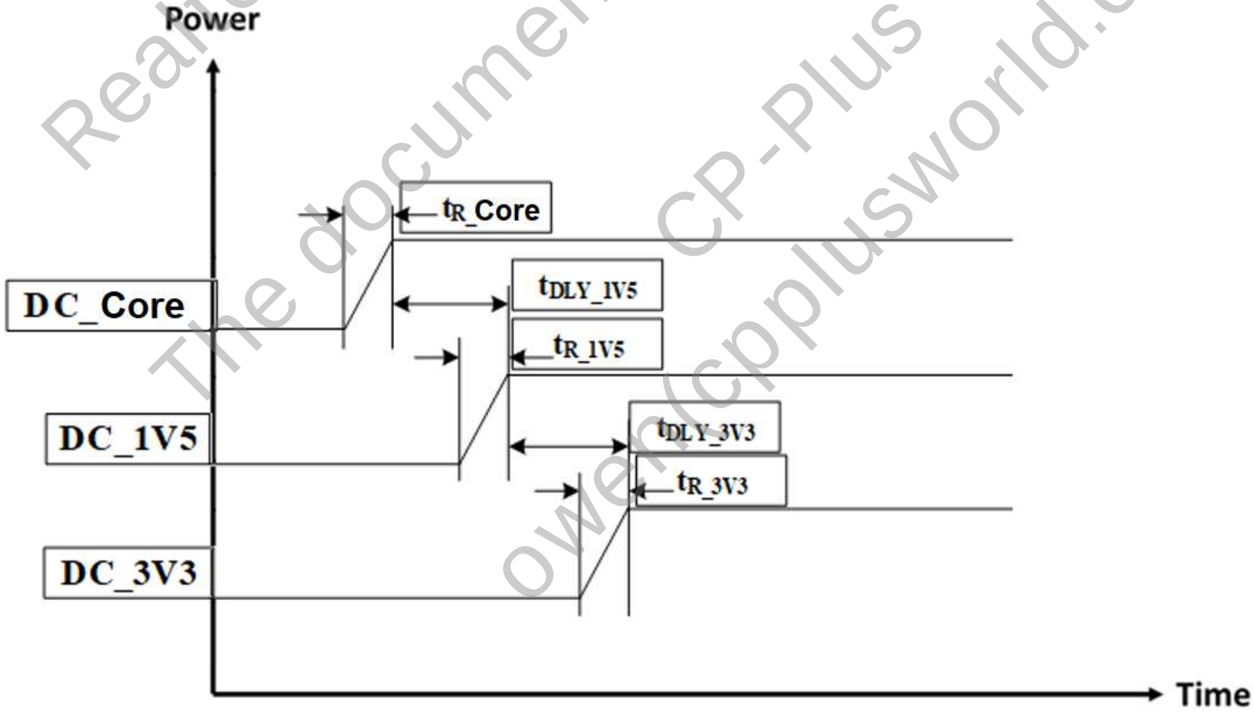


Figure 10-1 RTS3917N Power On Sequence

Table 10-2 Power Requirements

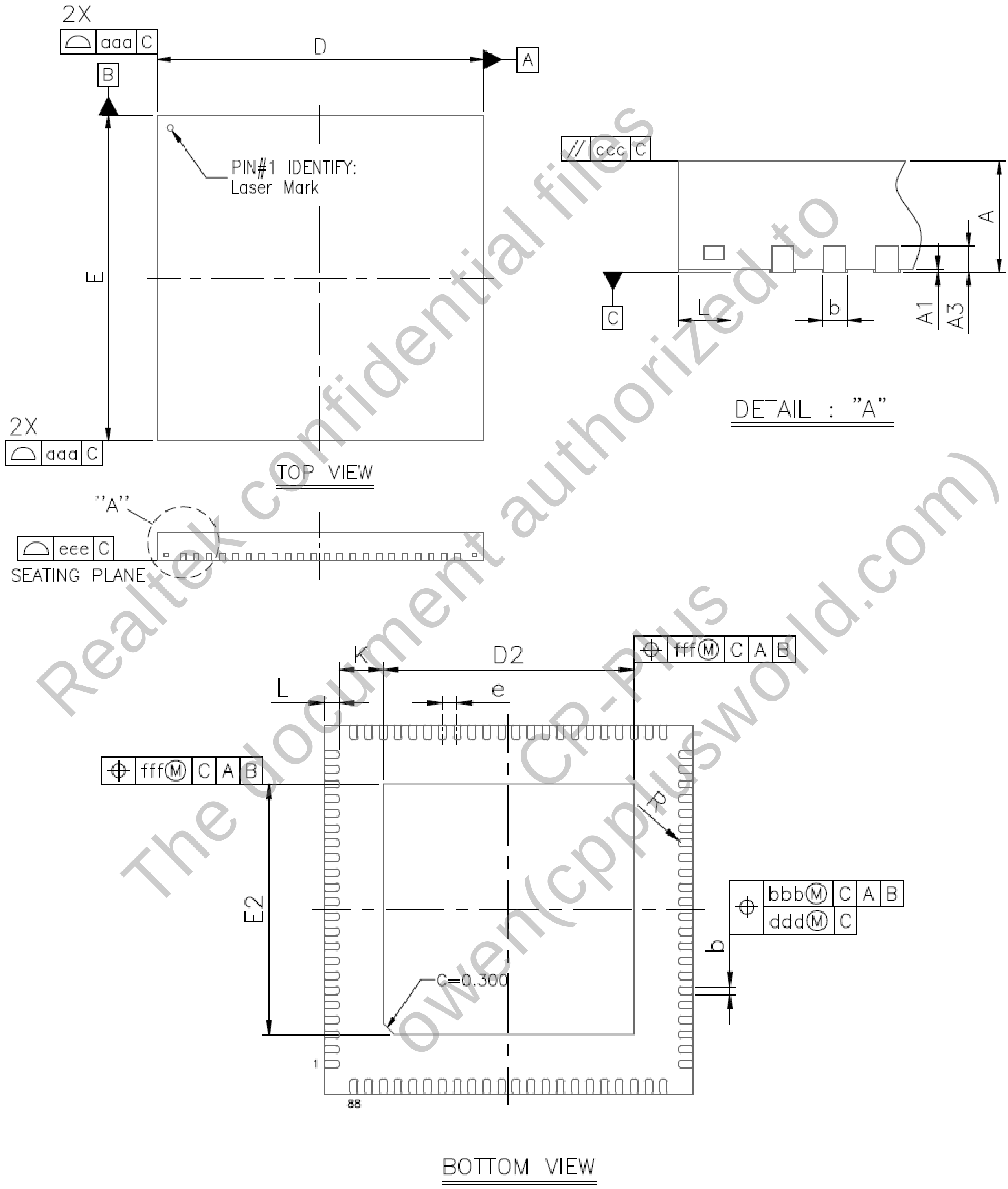
Name	Description	Min.	Max.	Unit
t_R_Core	DC_Core Rise Time	0	5	ms
t_R_1V5	DC_1V5 Rise Time	0	5	ms
t_R_3V3	DC_3V3 Rise Time	0	5	ms
t_{DLY}_1V5	Delay between DC_Core ready and DC_1V5 ready	0	10	ms
t_{DLY}_3V3	Delay between DC_1V5 ready and DC_3V3 ready	0	10	ms

11. Application Circuits

To guarantee the best compatibility and image quality in hardware design with specific image sensor, please contact Realtek to get the up to date application circuits. Any modification is recommended to be double checked by Realtek. Realtek may update the latest application circuits without modifying this datasheet.

12. Package Mechanical Dimensions

12.1. *QFN88 Dimension*



Symbol	Dimension in mm			Dimension in inch		
	MIN	NOM	MAX	MIN	NOM	MAX
A	0.80	0.85	0.90	0.031	0.033	0.035
A1	0.00	0.02	0.05	0.000	0.001	0.002
A3	0.20 REF			0.008 REF		
b	0.15	0.20	0.25	0.006	0.008	0.010
D	9.90	10.00	10.10	0.390	0.394	0.398
E	9.90	10.00	10.10	0.390	0.394	0.398
D2	6.70	6.80	6.90	0.264	0.268	0.272
E2	6.70	6.80	6.90	0.264	0.268	0.272
e	0.40 BSC			0.016 BSC		
L	0.30	0.40	0.50	0.012	0.016	0.020
K	0.20	---	---	0.008	---	---
R	0.075	---	0.125	0.003	---	0.005
aaa	0.10			0.004		
bbb	0.07			0.003		
ccc	0.10			0.004		
ddd	0.05			0.002		
eee	0.08			0.003		
fff	0.10			0.004		

NOTE:

1. CONTROLLING DIMENSION : MILLIMETER

13. Ordering Information

Table 13-1. Ordering Information

Part Number	Package	Status
RTS3917N-GR	QFN88 10x10mm^2 'Green' Package	MP

Realtek Semiconductor Corp.
Headquarters
No. 2, Innovation Road II
Hsinchu Science Park, Hsinchu 300, Taiwan
Tel.: +886-3-578-0211. Fax: +886-3-577-6047
www.realtek.com